



Highly effective activation of peroxymonosulfate via oxygen-coordinated single-atom iron for water decontamination

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ABSTRACT

The traditional peroxymonosulfate-based advanced oxidation processes (PMS-AOPs) have been significantly limited by the high inputs of oxidants. Herein, an oxygen-coordinated Fe-based single-atom catalyst (Fe-N2O1/OCN) was developed as an efficient pathway to activate peroxymonosulfate (PMS). This catalyst allows for a sharp decrease in PMS dosage while avoiding the release of additional sulfate ions into the treated water. The PAPO generates bisphenol-A (BPA) radicals and $\text{SO}_4^{\cdot-}$ species adsorbed onto Fe-N2O1/OCN, triggering a chain reaction for BPA degradation through BPA-initiated PAPO. The excellent performance of this system is attributed to the oxygen-coordinated iron center, which weakens the coordination of PMS on Fe-N2O1 and prevents its pre-activation. Furthermore, this PMS activation pathway demonstrates high anti-interference performance and can be continuously conducted in complex water matrices for 20 days, providing a promising alternative to traditional PMS-AOPs.

1. Introduction

Clean, fresh water is essential for all forms of life. The rapid growth of industrial production has met the increasing demands of human society but has also resulted in various environmental issues, particularly water pollution [1,2]. Advanced oxidation processes (AOPs) are widely used in wastewater treatment due to their high efficiency and fast decomposition of organic contaminants [3–6]. However, AOP technologies based on radical reactions have limitations in real water applications. In traditional AOPs, the highly reactive species generated (such as sulfate radicals, hydroxyl radicals, or superoxide radicals) indiscriminately attack not only the target pollutants but also the catalyst surfaces,

as well as other substances present in the water, such as inorganic ions and natural organic matter (NOM) in real waters. This leads to excessive consumption of oxidants and the generation of toxic by-products [6–9].

To address these challenges, non-radical activated pathways using PMS as an oxidant have emerged as an alternative approach [10,11]. The non-radical activated pathways of PMS involve the generation of oxidative species like singlet oxygen ($^1\text{O}_2$), high-valent metals, and surface-activated PMS complexes. Among them, $^1\text{O}_2$ and high-valent metals have been shown extensively studied due to their better compatibility with NOM and inorganic ions [12,13]. However, their lifetimes are still short, with most of them being quenched before effectively reacting with organic pollutants. Consequently, a significant

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amount of PMS is unnecessarily consumed. This is mainly because these species convert PMS into short-lived species first, resulting in overuse [14]. In response to this challenge, a post-activation pathway of oxidant (PAPO) based on the formation of surface-activated PMS complexes has been developed. This approach involves adsorbing PMS onto the catalyst surfaces through nucleophilic interaction, forming stable oxidant@catalyst complexes that act as a robust oxidant for pollutant degradation. The advantage of this approach is that PMS is only consumed for the degradation of micropollutants, minimizing unnecessary consumption and reducing water treatment costs.

However, research on controlling the PAPO is still in its early stages. Currently, only a few metal oxides and carbon materials have been developed to obtain oxidant@catalyst complexes [8,15–20]. Most of the catalysts based on single atoms primarily undergo the pre-activation pathway of oxidant (Pre-APO), generating short-lived species rather than forming PMS@catalyst through the PAPO [21]. This is due to the strong coordination between the catalyst and PMS, which prolongs the bond length of the Fe-O or O-O, promoting their cleavage in the PMS@catalyst complex (Fig. 1A) [12,22]. Furthermore, the most recent reports indicate that the optimal utilization efficiency of PMS has reached 82 % [23].

Herein, inspired by metal oxides, we have introduced oxygen doping into graphitic carbon nitride to create a Fe-based single-atom catalyst named Fe-N2O1/OCN. Notably, the oxygen-coordinated single-atom iron center on Fe-N2O1/OCN weakens the coordination between Fe and

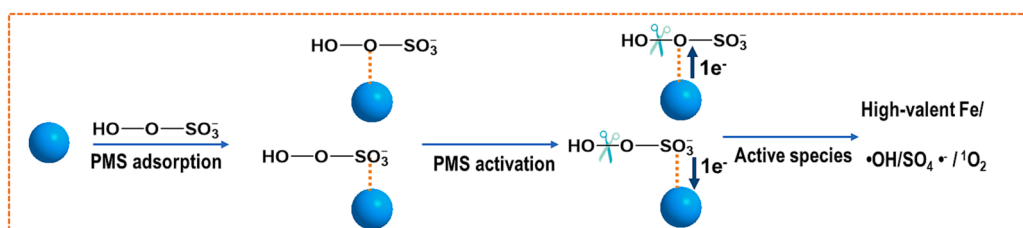
PMS. As a result, the O-O bond of PMS only cleaves when bisphenol A (BPA), a representative emerging contaminant, is present to undergo a PAPO reaction. Moreover, when activated by BPA, the PMS@Fe-N2O1/OCN simultaneously generates BPA radicals and $\text{SO}_4^{\bullet-}$ that adsorbed onto Fe-N2O1/OCN. These two species act as oxidants to initiate a chain reaction for BPA degradation, effectively reducing the consumption of PMS. In this activated pathway, Fe-N2O1/OCN achieves 150 % oxidant utilization, a significantly higher value compared to the reported highest one (82 %) among all materials in organic contaminant degradation systems. Furthermore, the Fe-N2O1/OCN system demonstrates continuous operation in complex water bodies for over 20 days without any decreases in performance. This PMS activation pathway provides a new approach to affordable, long-lasting, and interference-resistant AOP wastewater treatment.

2. Materials and methods

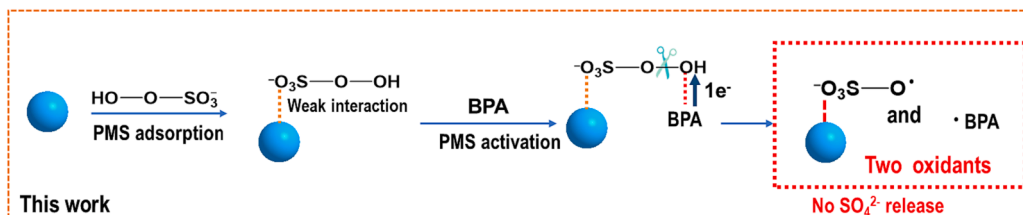
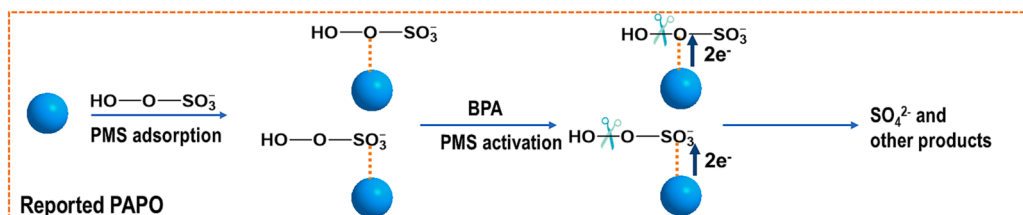
2.1. Material synthesis

Fe-based carbon catalyst (designated as Fe-N4/OCN) was synthesized using a simple method. Specifically, urea (15 g) was thoroughly mixed with oxalic acid (0.25 g) and $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ (0.1 g) in a mortar. The mixture was then transferred to a porcelain crucible with a lid and calcined in an air atmosphere at a heating rate of $5^\circ\text{C}/\text{min}$, maintaining a temperature of 500°C for 2 h. After naturally cooling to room

A Pre-activation pathways of oxidant (Pre-APO)



Post-activation pathways of oxidant (PAPO)



B

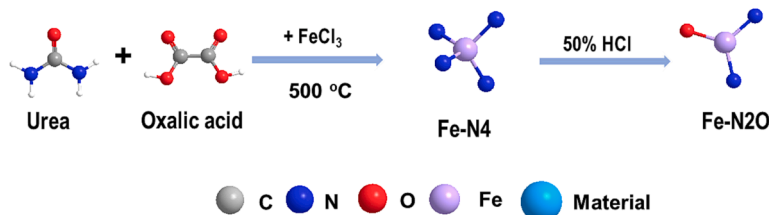


Fig. 1. The mechanism and the synthesis method of Fe-N2O1/OCN for achieving excellent performance. (A) Schematic representation of the preparation strategy, and (B) Synthetic procedure of Fe-N2O1/OCN.

temperature, a brown powder sample was obtained. The powder sample was washed five times with ultrapure water and three times with ethanol through centrifugation at 9000 rpm. The resulting solid samples were obtained by drying the suspension in an air dry oven at 80 °C for 24 h and subsequently stored at room temperature.

Fe-based single-atom catalyst (designated as Fe-N2O1/OCN) was synthesized by protonating Fe-N4/OCN. Specifically, Fe-N4/OCN (150 mg) was mixed with 4 mL of 6 M HCl and stirred at room temperature for 6 h. The sample was then washed 20 times with ultrapure water and three times with ethanol, followed by drying in a dry oven at 80 °C for 24 h. Yellow solid samples were obtained and stored at room temperature.

Fe-N2O1/OCN@ceramsite was synthesized using a similar method as Fe-N2O1/OCN, with the addition of commercial ceramsite.

2.2. Activity evaluation

The catalytic activity was evaluated in magnetically stirred conical flasks (100 mL) at room temperature (25 °C). In a typical experiment, a certain amount of PMS was added into 100 mL BPA (2 μM) solution to achieve the desired concentration (0.2 mM). The pH of the solution was adjusted to neutral (pH = 7 ± 0.2) by adding 0.1 M NaOH or 0.1 M H₂SO₄. The experiment was started after adding 10 mg catalyst to the sample. At predetermined time intervals, 1 mL of the reaction solution was taken out and immediately quenched with 0.1 mL of sodium thiosulfate (0.1 M). After that, the mixture solution was filtered by 0.45 μm PTFE hydrophilic filter and stored at 4 °C. Other micro-pollutants (PHBA, PCP, triclosan, PCA, MFA, 4-ABZ, aniline, and SMX) degradation in the PMS system was performed in a similar way as BPA. The kinetic rate constants (*k*_{obs}) of BPA degradation were further analyzed. The kinetics equation is as follows:

$$\ln \frac{C_t}{C_0} = -k_{obs}t \quad (1)$$

Certain amounts of scavengers such as MeOH (*k*'_{OH,MeOH} = 1.6–7.7 × 10⁷ M⁻¹s⁻¹, *k*'_{SO₄⁻,MeOH} = 1.2–2.8 × 10⁹ M⁻¹s⁻¹), TBA (*k*'_{OH,TBA} = 6.0 × 10⁸ M⁻¹s⁻¹, *k*'_{SO₄⁻,TBA} = (4–9.5) × 10⁵ M⁻¹s⁻¹), and FFA (*k*'_{FFA, ¹O₂} = 1.2 × 10⁸ M⁻¹s⁻¹), were used to distinguish the role of •OH, SO₄•⁻ and ¹O₂, respectively. SOD, Catalase, K₂Cr₂O₄, and NaOA were used to capture superoxide radical (O₂•⁻), H₂O₂, free electrons, and holes, respectively. DMSO was used as the quencher for high-valent iron-oxo species. In addition, for the experiments studying the effect of molecular oxygen on BPA degradation, the reaction solutions were bubbled by O₂ or Ar for 90 min before the reaction started, respectively. The real water was collected from Beijiang River, secondary effluent, and landfill leachate, and BPA was added to the real water to reach a final concentration of 2 μM. All the degradation experiments were conducted in duplicate to ensure accuracy. The mean value and standard deviations are calculated and recorded.

The modified rate constant (*k*-value) is calculated as Eq. (2),

$$k - \text{value} = \frac{k_{obs}[\text{pollutant}]}{[\text{catalyst dosage}][\text{oxidant dosage}]} \quad (2)$$

The PMS utilization efficiency is calculated as Eq. (3),

$$\text{PMS utilization} = \frac{[\text{BPA degraded}]}{[\text{PMS consumed}]} \times 100\% \quad (3)$$

2.3. Analytical methods

The concentration of micropollutants was analyzed on the high-performance liquid chromatography (HPLC, UltiMateTM 3000, Thermo Fisher Scientific Inc., USA) equipped with a CNW Athena C18-WP column (110A, 4.6 × 150 nm, 5 μm, ANPEL Laboratory Technologies), a diode array detector and a fluorescence detector. The HPLC

method for each micropollutant is shown in Table S1. The dissolved organic carbon (DOC) of water samples was analyzed using a TOC analyzer (TOC-L CPH). Before being analyzed, the water sample should be filtered by a 0.45 μm hydrophilic filter. The concentration of PMS was analyzed by the iodometric titration method using a spectrophotometer (Shimadzu UV-1800, Japan) at the wavelength of 352 nm. The concentration of total Fe was measured by ICP-MS (Agilent 7700(MS)).

The relative contribution of ¹O₂ on BPA degradation removal was evaluated by a modified method [24]. [¹O₂]_{ss} presents the steady-state concentration of ¹O₂, which was calculated using Eq. (4),

$$[\text{}^1\text{O}_2]_{ss} = \frac{k_{obs,FFA}}{k'_{FFA, \text{}^1\text{O}_2}} \quad (4)$$

where *k*'_{FFA, ¹O₂} represents the second-order rate constants of ¹O₂ reacting with FFA (as the probe compound of ¹O₂, *k*'_{FFA, ¹O₂} = 1.2 × 10⁸ M⁻¹s⁻¹). *k*_{obs,FFA} represents the first-order rate constants of ¹O₂ reacting with FFA.

$$k_{\text{}^1\text{O}_2, \text{BPA}} = k'_{\text{}^1\text{O}_2, \text{BPA}} [\text{}^1\text{O}_2]_{ss} \quad (5)$$

$$\eta = \frac{k_{\text{}^1\text{O}_2, \text{BPA}}}{k_{obs, \text{BPA}}} \quad (6)$$

where *k*'_{¹O₂, BPA} represents the second-order rate constant of ¹O₂ with BPA (*k*'_{¹O₂, BPA} = 8 × 10⁵ M⁻¹s⁻¹). *k*_{¹O₂, BPA} represents the first-order rate constant of ¹O₂ with BPA. *k*_{obs, BPA} represents the observed first-order rate constant of BPA in the reaction system. η is the contribution ratio of ¹O₂ in the PMS system.

2.4. Electrochemical experiments

All electrochemical measurements were conducted in a three-electrode cell using an electrochemical workstation (CHI6009E, China). The activator-coated GCE, Ag/AgCl/KCl, and platinum electrodes acted as the working, reference, and counter electrode, respectively. A 0.1 M phosphate buffer (pH = 7 ± 0.1) was used as the electrolyte. The activator-coated GCE was prepared by drying samples suspension on the GCE. The sample suspension was prepared by mixing 10 mg sample with 0.5 mL ethanol and 50 μL of Nafion perfluorinated resin solution, followed by sonication for 1 h. For cyclic voltammetry (CV), the potential was swept from 0.2 V to 0.3 V (vs. Ag/AgCl). For current-time curves, initial potential was set at 0.4 V and PMS was added into the reactor at 100 s with final concentrations of 2 mM. The BPA solution was added into the reactor at 200 s with a final concentration of 20 μM, and the reactor was stirred at 550 r/min throughout the process.

2.5. Theoretical calculations

All density functional theory (DFT) calculations were performed by using the Vienna ab initio Simulation Program (VASP) [25,26]. The calculations adopted the generalized gradient approximation (GGA) in the Perdew-Burke-Ernzerhof (PBE) form, with a cutoff energy of 500 eV for a plane-wave basis set [27]. A 5 × 5 × 1 Monkhorst-Pack grid was used to sample the Brillouin zones during structure optimization [28]. Ion-electron interactions were described using the projector augmented wave (PAW) method [29]. Convergence criteria for the structure optimization were set such that the maximum force on each atom was less than 0.01 eV/Å and the energy change was less than 1 × 10⁻⁵ eV. The DFT-D3 semiempirical correction, based on Grimme's scheme, was included [30]. The Gibbs free energy change (ΔG) for each elemental step is defined as [31,32]:

$$\Delta G = \Delta E + \Delta ZPE - T\Delta S + \Delta G_U + \Delta G_{pH} \quad (7)$$

where ΔE represents the adsorption energy based on DFT calculations,

ΔZPE is the zero-point energy correction, T is the temperature, ΔS is the entropy change, U is the applied electrode potential, and ΔG_{pH} is the free energy correction of the pH.

3. Results and discussion

3.1. Synthesis and structural characterization

The synthesis procedures of Fe-N2O1/OCN and Fe-N4/OCN are illustrated in Fig. 1B. We initially synthesized a series of Fe catalysts based on graphitic carbon nitride using a simple one-step calcination method with $FeCl_3$, urea, and oxalic acid as the precursors. For protonation, the Fe-N4/OCN samples were further washed with 6 M hydrochloric acid. The resulting protonated catalyst was named Fe-N2O1/OCN. Control samples including pristine polymetric carbon nitride (referred to as CN) and oxygen-doped CN (referred to as OCN), were also prepared using similar procedures [33].

The crystalline structures and structural compositions of both Fe-N4/OCN and Fe-N2O1/OCN samples were examined using powder X-ray diffraction (XRD) (Fig. S1) and Fourier transform infrared (FTIR) (Fig. S2). The results showed that their structures were similar to pristine CN, indicating the successful polymerization of graphite carbon nitride. Additionally, high-resolution transmission electron microscopy (HR-TEM) images (Fig. S3-S4) revealed that both Fe-N4/OCN and Fe-N2O1/OCN exhibited porous lamellar structures. No Fe nanostructures were observed in either catalyst. Aberration-corrected high-angle annular dark field scanning transmission electron microscopy (HAADF-

STEM) (Fig. 2A) was conducted to investigate the isolation feature of iron atoms in Fe-N2O1/OCN. Bright spots corresponding to high-density Fe single atoms were observed throughout the carbon nitride matrix, marked with yellow circles. The element mapping of Fe-N2O1/OCN further confirmed the uniform distribution of atomically dispersed Fe atoms on the carbon nitride nanosheets.

XPS analysis revealed a noticeable shift in elemental composition when comparing Fe/CN (synthesized without oxalic acid using $FeCl_3$ and urea as precursors) to Fe-N2O1/OCN, as detailed in Table S4. Specifically, there was an increase in oxygen content while the nitrogen content decreased in Fe-N2O1/OCN. Additionally, in the high-resolution XPS analysis, Fe-N2O1/OCN exhibited absorption peaks at 531.5 eV (attributed to C-O-C) and 286.4 eV (corresponding to O-C-O) in the peak-fitted O 1s and C 1s regions [34]. These findings strongly indicated the successful substitution of oxygen atoms for nitrogen atoms, with the incorporated oxygen manifested as C-O-C groups within the OCN substrate in Fe-N2O1/OCN.

X-ray absorption spectroscopy (XAS) was conducted to determine the coordination of Fe atoms in different environments. The Fourier-transformed extended X-ray absorption fine structure (FT-EXAFS) spectra (Fig. 2C) exhibited one main peak at $\approx 1.54 \text{ \AA}$ for Fe-N4/OCN, indicating Fe-N coordination. The coordination number of Fe-N in Fe-N4/OCN was determined to be 4, with a bond length of $2.04 \text{ \AA}$, based on the FT-EXAFS data fitting in R space (Table S2). For Fe-N2O1/OCN, the main peak was slightly broadened and shifted to $\approx 1.67 \text{ \AA}$ (indicating Fe-N or Fe-O coordination), indicating a distinct coordinated environment compared to Fe-N4/OCN. Additionally, the diminished

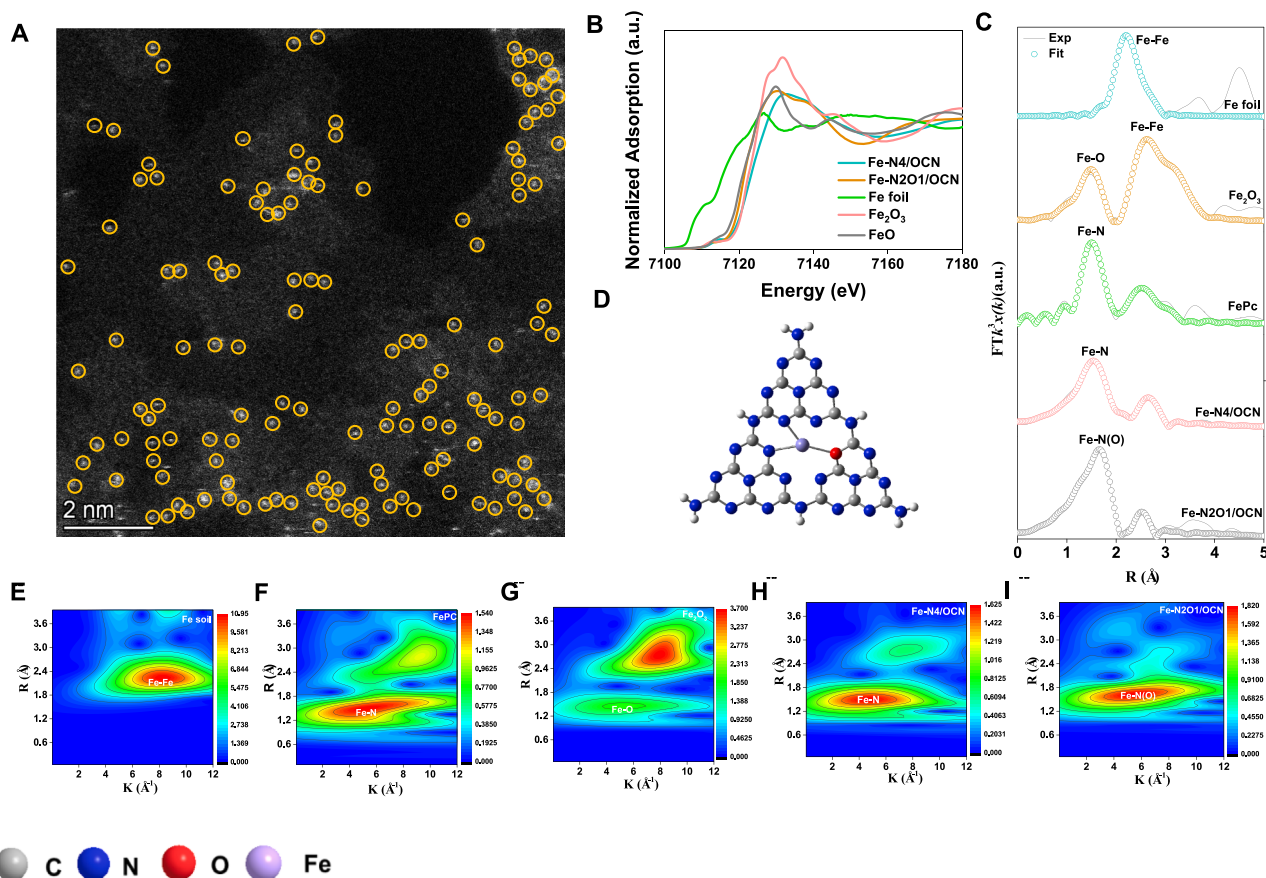


Fig. 2. Characterizations of Fe-N2O1/OCN catalyst. (A) AC-HAADF-STEM image of Fe-N2O1/OCN (the Fe atoms are marked by yellow circles). (B) Fe XANES spectra. (C) EXAFS in R-space and corresponding fitting curves and (E-I) WT of EXAFS for Fe foil, Fe_2O_3 , FePc, Fe-N4/OCN, and Fe-N2O1/OCN, respectively. (D) The revised structure of the Fe-N2O1 site in Fe-N2O1/OCN (● C ● N ● O ● Fe). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

intensity of the pre-edge peak and the increased intensity of the white-line peak in the case of Fe-N2O1/OCN (Fig. S5) indicated a decrease in the symmetry of the Fe coordination environment [35]. Moreover, the central position of Fe-N2O1/OCN demonstrated a positive shift relative to Fe-N4/OCN in the wavelet transforms (WT)-EXAFS spectra (Fig. S5), indicating a potential coordination of the Fe atoms with a larger atom, such as O, in this study. Based on the FT-EXAFS data fitting in R space (Table S2), for Fe-N2O1/OCN, the coordination numbers were 1 for Fe-O and 2 for Fe-N, with bond lengths of 1.90 Å and 2.11 Å, respectively.

Moreover, theoretical modeling was conducted to identify potential oxygen-doped sites for coupling with the Fe-N2O1 center on the OCN surface in the structures of Fe-N2O1/OCN. Based on the analysis before, two possible oxygen-doped sites were considered (Fig. S9). The analysis revealed that the oxygen within the OCN was doped on the nitrogen atom at the edge of the heptazine ring, rather than the nitrogen atom linking the two heptazines (Fig. S9). The optimized structures revealed that the coordination environment of Fe in Fe-N2O1/OCN consisted of two nitrogen atoms and one oxygen atom, which was consistent with the results obtained from the EXAFS analysis.

The oxidation states of Fe-N4/OCN and Fe-N2O1/OCN were also determined. The position of the rising edge in X-ray absorption near-edge structure (XANES) spectra for Fe-N4/OCN and Fe-N2O1/OCN (Fig. 2B) fell between those of FeO and Fe₂O₃, indicating the presence of single Fe atoms with a positive charge and an average oxidation state between Fe²⁺ and Fe³⁺. In comparison, the valence of single Fe atoms in Fe-N2O1/OCN was closer to +2 than that of Fe-N4/OCN. It was observed that the OCN substrate in Fe-N2O1/OCN was partially protonated through acid treatment. The zeta potential values of protonated OCN and Fe-N2O1/OCN dispersed in DI water (+18.36 mV and +16.46 mV, respectively) were higher than those of OCN and Fe-N4/OCN (+4.16 mV and +3.06 mV, respectively) (Fig. S8).

3.2. Evaluation of catalytic activity

The catalytic activity trends of g-C₃N₄-supported catalysts were

further evaluated using the BPA degradation process. As shown in Fig. S11, negligible BPA (<2%) can be adsorbed by Fe-N2O1/OCN within 15 min. Fig. 3A shows the results based on the first-order kinetic constants (k_{obs}). The Fe-doped carbon nitride catalysts all demonstrated rapid BPA degradation, following the sequence of Fe-N2O1/OCN > Fe-N4/OCN > OCN > CN. Among them, Fe-N2O1/OCN exhibited the highest catalytic performance with a k -value as high as 235.95 min⁻¹ M⁻¹ in the presence of 42.9 mg/L BPA, surpassing most state-of-the-art heterogeneous catalysts and single-atom catalysts (Fig. S12, Table S6, and Table S7). The impact of the initial concentration of BPA (ranging from 2 μM to 100 μM) is illustrated in Fig. S13. As anticipated, the degradation efficiency gradually decreased with increasing initial concentration of BPA. However, when the BPA concentration was below 100 μM, over 90 % of BPA could still be removed within 15 min. Remarkably, Fe-N2O1/OCN exhibited much higher activity compared to Fe-N4/OCN and the individual OCN, as well as physically mixed FeCl₃ and OCN (Fig. S14), indicating that the coordination environment of Fe significantly influenced its catalytic performances. Moreover, the accelerated pollutant removal by the Fe-N2O1/OCN system compared to Fe-N2O1/OCN or PMS alone confirmed the dominance of the AOPs catalytic process.

Impressively, even when the original concentration of PMS approached the concentration of BPA, Fe-N2O1/OCN was still able to degrade BPA at a rate of 0.11 min⁻¹ (Fig. S15-S16 and Fig. 3B) with the oxidant utilization efficiency of 150 % (i.e., [BPA degraded]/[PMS consumed] × 100 %), which is 2.4 times larger than that in the Fe-N4/OCN system (Fig. S17) and is the highest oxidant utilization among all reported materials. Additionally, Fe-N2O1/OCN maintained similar degradation rates for a wide range of organic pollutants (Fig. S18), and the k_{obs} of these pollutants showed a well-defined linear correlation ($R^2 = 0.83$) with half-wave potential ($\phi_{1/2}$) (Fig. S19). Furthermore, Fe-N2O1/OCN also exhibited activation of low concentration of H₂O₂ (0.2 mM) and achieved 70 % BPA removal within 15 min (Fig. S20), demonstrating its broad catalytic activity.

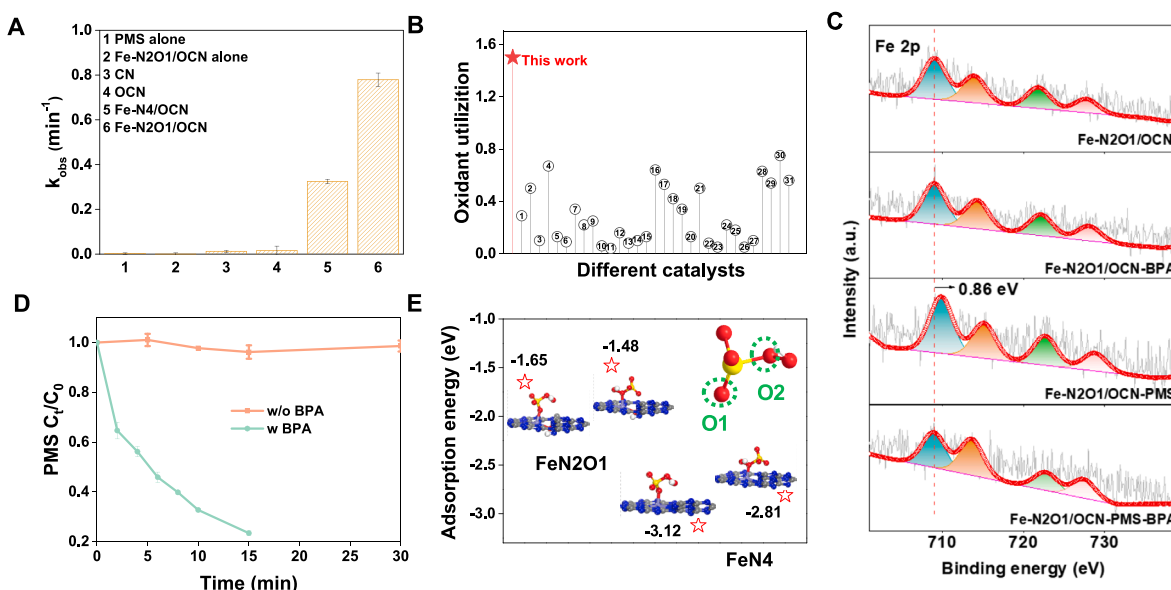


Fig. 3. Catalytic performances of several materials in PMS-assisted systems. (A) BPA degradation rate in different catalysts-activated PMS systems. Reaction condition: [BPA] = 2 μM, [catalyst] = 0.1 g/L, [PMS] = 0.2 mM, initial solution pH 7.0 ± 0.2. (B) The oxidant utilization of Fe-N2O1/OCN and other recently reported catalysts. (1 Cu1/NG, 2 CNTs, 3 CNT, 4 Cu-N4/C-B, 5 Cu-N4/C-P, 6 ZnFe-LDH, 7 ZnFeMnO₄, 8 Fe/O-doped g-C₃N₄, 9 Fe(III)-doped g-C₃N₄, 10 PEI-MWCNT, 11 Co-N₃O₁₋₁, 12 CoN₂₊₂, 13 Fe-SAC, 14 Fe1/C, 15 FeCoNC, 16 AND/800, 17O-CNTs-1000@PMS, 18 Fe_{0.15}Mn_{0.85}O, 19 CPPy-F-8, 20 CuO, 21 Co₃O₄, 22 FeSAC, 23 Fe-Co-O-g-C₃N₄, 24 Co₃O₄/CeO₂-IM, 25p-CoSi1N₃@D, 26 Cu-Ln₂O₃/Ov, 27 Co-BO, 28 SAZn-NC, 29 A-CoMnO, 30 A/C-CoMnO, 31 Co₃@Mn_xO₄). (C) The XPS Fe 2p spectra of Fe-N2O1/OCN before and after reaction with BPA, PMS, and BPA/PMS, respectively. (D) PMS decomposition in Fe-N2O1/OCN system with or without BPA. Experimental conditions: [PMS] = 0.2 mM, [catalyst] = 0.1 g/L, [BPA] = 40 μM, initial solution pH 7.0 ± 0.2. (E) The adsorption energies of different oxygen sites of PMS onto Fe-N2O1/OCN and Fe-N4/CN.

3.3. Reactive species and degradation process

The scavengers using MeOH (20 mM), TBA (20 mM), and superoxide dismutase (SOD, 50 U/mL) did not influence the degradation of BPA in the Fe-N2O1/OCN activated PMS system (Fig. S21). Benzoic acid (BA), which is reactive with $\text{HO}^\bullet$ ($4.2 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$) and $\text{SO}_4^{\bullet-}$ ($1.2 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$) [36] also showed no degradation (Fig. S22). The evidence implied the negligible formation of $\text{SO}_4^{\bullet-}$, $\text{HO}^\bullet$, and $\text{O}_2^{\bullet-}$ in the system. The addition of dimethyl sulfoxide (DMSO), a quencher for Fe(VI)/Fe(V) , had no negative effects on BPA degradation, which excluded the role of high-valent iron species in the system (Fig. S23). The presence of DMSO demonstrated a slight enhancement in BPA degradation, potentially attributed to the possible generation of organic radicals from DMSO in the system [37], which may react with BPA.

When 5,5-dimethyl-1-pyrroline-N-oxide (DMPO) and 2,2,6,6-tetramethyl-4-piperidinyloxy (TEMP) were employed as trapping reagents, the EPR results indicated the absence of signals for $\text{DMPO-SO}_4^{\bullet-}$ and $\text{DMPO-HO}^\bullet$ in the Fe-N2O1/OCN activated PMS system (Fig. S24). The results were consistent with the quencher tests. The characteristic signals of $\text{TEMP-}^1\text{O}_2$ were identified instead (Fig. S25). In order to further verify the role of $^1\text{O}_2$ in BPA degradation, the tests were conducted using D_2O , in which the lifetime of $^1\text{O}_2$ ($68 \pm 1 \mu\text{s}$) is 18 times longer than in H_2O ($3.7 \pm 0.4 \mu\text{s}$) [38,39]. BPA degradation should be accelerated if $^1\text{O}_2$ is a major oxidant. In fact, the changing of solvent did not accelerate the degradation rate of BPA (Fig. S26). To make further quantification of

$^1\text{O}_2$, the degradation of furfuryl alcohol (FFA), which is very reactive with $^1\text{O}_2$ ($k' = 1.2 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$ at pH 7.0), was tracked in PMS alone, and the Fe-N2O1/OCN activated PMS system (Fig. S27). The observed decay rate of FFA was 0.0292 min^{-1} . As the secondary-reaction rate constant of BPA with $^1\text{O}_2$ ($k = 8 \times 10^5 \text{ M}^{-1}\text{s}^{-1}$ at pH 7.0) [40] was 150 times lower than that of FFA, the contribution of $^1\text{O}_2$ in BPA degradation was calculated to be less than 0.02 % in the Fe-N2O1/OCN activated PMS system. As such, the degradation of BPA was not attributed to $\text{SO}_4^{\bullet-}$, $\text{HO}^\bullet$, $\text{O}_2^{\bullet-}$, $^1\text{O}_2$, and high-valent iron in the Fe-N2O1/OCN activated PMS system.

Sodium oxalate (NaOA) is a chemical that can scavenge the holes on the catalyst surface [41]. BPA was not inhibited with NaOA addition (Fig. S21), suggesting the negligible role of holes of catalyst if present. Electron transfer could be the possible reaction pathway resulting in BPA degradation. Then potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) was used as a scavenger for the solution-phase electrons [14]. However, it had no effects on BPA degradation (Fig. S21), indicating the absence or void effects of aqueous electrons.

Subsequently, we investigated the possibility of PAPO. Specifically, we found that PMS was not decomposed by the catalyst and only consumed in the presence of contaminants. In our system, when PMS and Fe-N2O1/OCN were present, there was negligible decomposition of PMS. However, the decomposition of PMS was sustained after the addition of contaminants to the system (Fig. 3D and Fig. S28). More than 80 % PMS decomposed in the presence of $40 \mu\text{M}$ BPA within 15 min. The

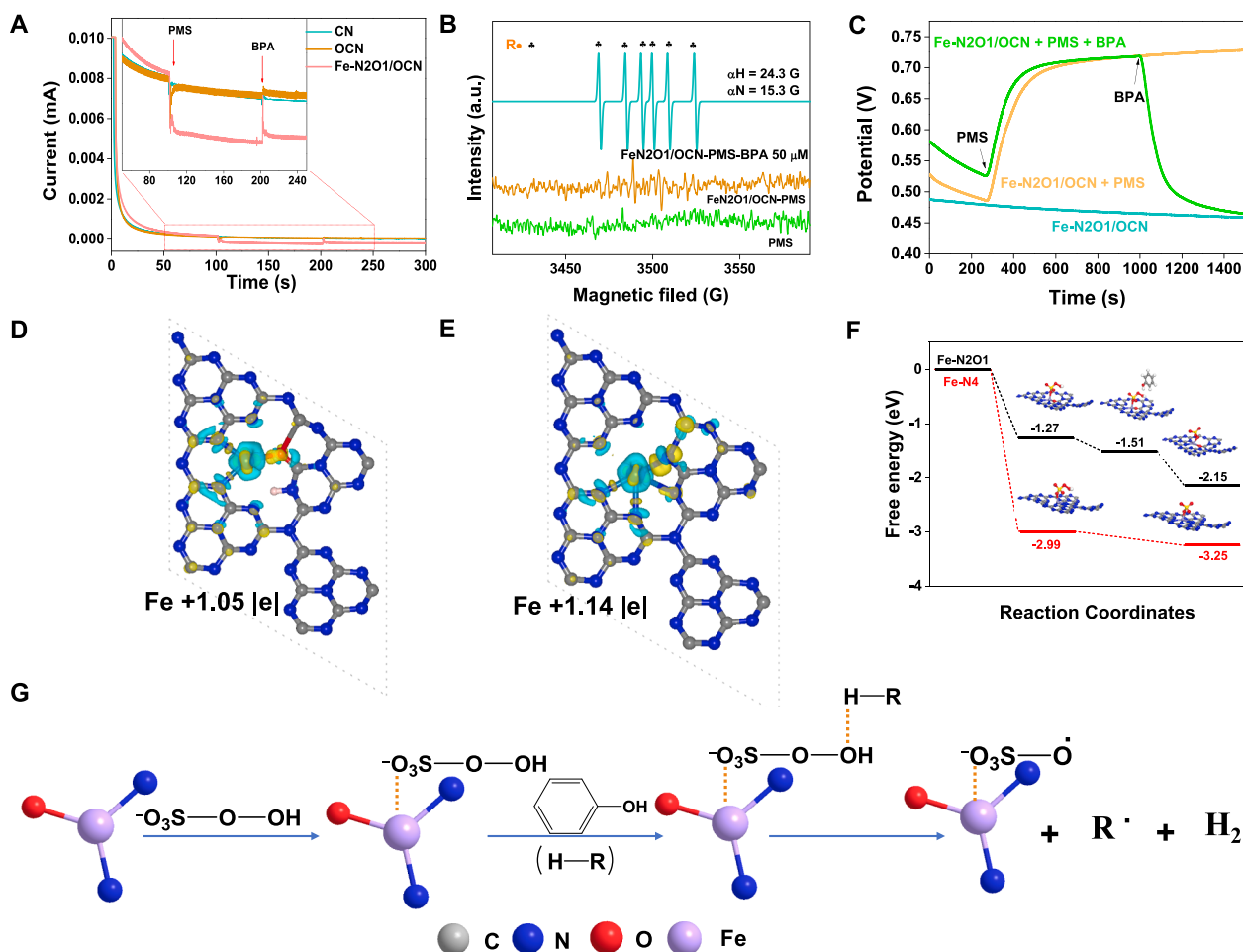


Fig. 4. Unraveling the mechanism of the reaction on Fe-N2O1/OCN catalyst. (A) Amperometric *i-t* curve measurements using CN, OCN, and Fe-N2O1/OCN as the working electrode, respectively. (B) EPR spectra of generated $\text{R}^\bullet$ in Fe-N2O1/OCN-PMS-BPA systems. This experiment was conducted by combining PMS, Fe-N2O1/OCN, and BPA and reacting for 5 min. Before detecting the EPR signal, the samples should be filtered and combined with DMPO. (C) The open-circuit potential of Fe-N2O1/OCN as the working electrode. The optimized structure and the corresponding electron density plots of (D) Fe-N2O1/OCN and (E) Fe-N4/CN. (F) The free energies for PMS activation on the Fe-N2O1 and Fe-N4 sites. (G) The proposed reaction pathway of PMS activation by Fe-N2O1/OCN.

result indicated that the Fe-N2O1/OCN system was in a pathway of PAPO.

Furthermore, the Fe 2p XPS of Fe-N2O1/OCN showed an increase of 0.82 eV after the addition of PMS (Fig. 3C), indicating the formation of PMS@Fe-N2O1/OCN complex. It suggests that PMS may be absorbed by the Fe atom of Fe-N2O1/OCN. This finding was also supported by DFT calculation. The optimized structure revealed that PMS weakly chemisorbs to the Fe atom ($E_{ad} = -1.65$ eV). Interestingly, the O1 site of PMS was preferred for absorption on the Fe atom, rather than the conventional O2 site, promoting the breakage of the O-O bond in PMS (Fig. 3E and Fig. S29).

The formation of the PMS@Fe-N2O1/OCN complex was further supported by the open-circuit potential. As shown in Fig. 4C, the open-circuit potential on the catalyst electrode increased upon the addition of PMS and then reached a plateau, indicating the formation of PMS@Fe-N2O1/OCN complex. Moreover, the Fe 2p XPS of Fe-N2O1/OCN remained unchanged after the addition of BPA (Fig. 3C), indicating weak absorption of BPA. Consistently, negligible adsorption of BPA was detected (Fig. 3A). The weak adsorption of BPA hinders its oxidation through the interfacial electron transfer pathway. This finding was also supported by the perchlorate test, which distinguishes between the interfacial electron transfer pathway (inner-sphere) and the direct activation pathway of PMS@Fe-N2O1/OCN (outer-sphere) [19]. The perchlorate test had a negative effect on the Fe-N2O1/OCN system, confirming the direct activation pathway of PMS@Fe-N2O1/OCN (Fig. S30).

Moreover, the amperometry i-t curves of the electrochemical system with catalysts loading on the working electrode were used to determine the direction of electron flow (Fig. 4A). Specifically, a sudden decrease in current was observed after the addition of PMS, followed by a partial rebound in current after injection of BPA. These results indicated the electron cloud of Fe-N2O1/OCN was reduced after the formation of PMS@Fe-N2O1/OCN. Additionally, the peak of Fe 2p XPS increased by 0.82 eV after the formation of PMS@Fe-N2O1/OCN, which corresponded to the lower electron cloud of Fe after the addition of PMS. The process of partial current rebounding after the injection of BPA was also investigated.

It is important to note that the BPA radical ($BPA^{\bullet}$), formed by the simultaneous loss of one electron and one proton from BPA simultaneously, was detected using EPR. The peak shapes were consistent with the DMPO- $R^{\bullet}$ [42], which $R^{\bullet}$ is attributed to the unsaturation carbon-centered radical (Fig. 4B). Furthermore, a polymerization product was detected in the bulk solution (Fig. S31), providing supporting evidence for the existence of $BPA^{\bullet}$ [43]. Therefore, the degradation of BPA by the PMS@Fe-N2O1/OCN complex occurs through a proton-coupled electron-transfer pathway. This process is stoichiometric in theory, resulting in a substantial reduction in the amount of PMS. The presence of $R^{\bullet}$ also indicated the O-O bond in PMS@Fe-N2O1/OCN was homolytic cleavage, and the other product of this cleavage could be the adsorption of $SO_4^{\bullet-}$ onto Fe-N2O1/OCN. Both $R^{\bullet}$ and $SO_4^{\bullet-}$ adsorbed on Fe-N2O1/OCN act as the oxidants for BPA degradation, leading to the production of polymerization products and organic sulfides (Fig. S31-S32, Table S10). Note that, the LC50 or EC50 of organic sulfides was larger than that of BPA in fish, daphnia and green algae. The proposed degradation pathway for BPA in the system is shown in Fig. S33. The degradation of BPA was mainly through polymerization, hydroxylation, ring elimination, electronic rearrangement, sulfation, oxidation, and ring opening [44-49]. P1, resulting from the BPA radical ($BPA^{\bullet}$), represented the polymerization product. P2 and P3 were formed via a series of hydroxylation. Subsequently, P4 was generated through ring elimination and electronic rearrangement of P3. P8 was the product of oxidation and ring opening. P5 was produced by ring elimination and hydroxylation on BPA. The formation of P6 occurred through a combination of P5 and M1. P7 was generated by sulfation on P6. It is worth noting that other reported PMS activation catalysts only generated organic radicals for further degradation of organic pollutants [50].

To further demonstrate the ability of the PMS@Fe-N2O1/OCN complex in directly oxidizing BPA, open-circuit potentials were measured (Fig. 4C), and found to be at 0.7 V (vs Ag/AgCl), surpassing the oxidation potential of BPA (0.46 V, vs Ag/AgCl), as indicated by its half-wave potential of BPA [51].

Additionally, in-situ Raman spectroscopy demonstrated the presence of a new peak at 832 cm^{-1} in the system including Fe-N2O1/OCN and PMS (Fig. S34). This characteristic peak can be attributed to the coordination between PMS and Fe, confirming the existence of PMS@Fe-N2O1/OCN [23,52].

3.4. The mechanism for the activation pathway of post-oxidant

The regulation mechanism of oxidative species through the coordination environment was investigated using DFT calculations. In comparison to the conventional coordination mode of one Fe atom coordinated with four nitrogen atoms (Fe-N4), which is commonly observed in iron single-atom catalysts with carbon nitride as the substrate, the oxygen-coordinated Fe-N2O1 coordination environment led to a decrease in the electron cloud density of the iron center (Fig. 4D-4E). As a result, the coordination energy between PMS and Fe-N2O1/OCN was significantly weakened, causing the iron center to prefer the more electronegative O1 coordination in PMS over O2 (1.65 eV for O1, and 1.48 eV for O2) (Fig. 3E), resulting in the formation of PMS@Fe-N2O1/OCN. Additionally, the electron flow in Fe-N2O1/OCN was directed from Fe to PMS, which is opposite to that in Fe-N4 [12]. The weak coordination mediated the absorbed sites of PMS and the direction of electron flow, effectively preventing stretching of the Fe-O or O-O bond in PMS@Fe-N2O1/OCN, thereby promoting the PAPO.

The Gibbs free energy of the reaction was further calculated to explore the activation mechanism, as illustrated in Fig. 4F, depicting the intermediate structures and their corresponding energy distributions. Activation of PMS in Fe-N4 sites and Fe-N2O1 sites is both thermodynamically spontaneous. The more negative adsorption-free energy on the Fe-N4 sites demonstrated a higher PMS adsorption efficiency compared to the Fe-N2O1 site. In the PMS dissociation step, the dissociation energy at Fe-N4 is -0.26 eV, which is much smaller than the dissociation energy of PMS at the Fe-N2O1 site. This suggests that the O-O bond breaks more easily at Fe-N4 sites than at the Fe-N2O1 sites. Regarding Fe-N2O1, the pollutant approaching the complex and subsequent adsorption of $SO_4^{\bullet-}$ onto Fe-N2O1 aligns with thermodynamic feasibility. This information underscores the favorable conditions for the adsorption process on Fe-N2O1 sites, emphasizing its potential as an effective site for PMS activation in the real water and pollutant removal.

Generally, PMS spontaneously adsorbed onto the Fe sites of the Fe-N2O1/OCN, forming PMS@Fe-N2O1/OCN. Subsequently, the pollutant (for convenience, phenol was used as a substitute for BPA) approached the PMS@Fe-N2O1/OCN through weak interaction. Finally, the O-O bond in PMS@Fe-N2O1/OCN underwent homolytic cleavage, generating $BPA^{\bullet}$, $SO_4^{\bullet-}$ adsorbed onto Fe-N2O1/OCN and H_2O . The $BPA^{\bullet}$, $SO_4^{\bullet-}$ adsorbed onto Fe-N2O1/OCN initiated a chain reaction for BPA elimination.

3.5. Evaluation of Fe-N2O1/OCN in practical decontamination application

The degradation rate of BPA was not significantly affected by the presence of cations (e.g., Mg^{2+} , Ca^{2+} , K^+ , Na^+), anions (e.g., Cl^- , SO_4^{2-} , HPO_4^-), and NOM (Fig. 5A). Although the presence of HCO_3^- and NO_3^- had minor inhibitory effects on BPA degradation, the degradation rate of BPA remained high and it could be eliminated within 8 min (Fig. S35A). Furthermore, the degradation of BPA showed minimal variation over a wide pH range from 3 to 11 (Fig. 5B). Additionally, the system was unaffected by dissolved oxygen concentrations as the purging of the solution with air, oxygen, and argon did not inhibit BPA degradation (Fig. S35B). According to the preceding results, the Fe-N2O1/OCN

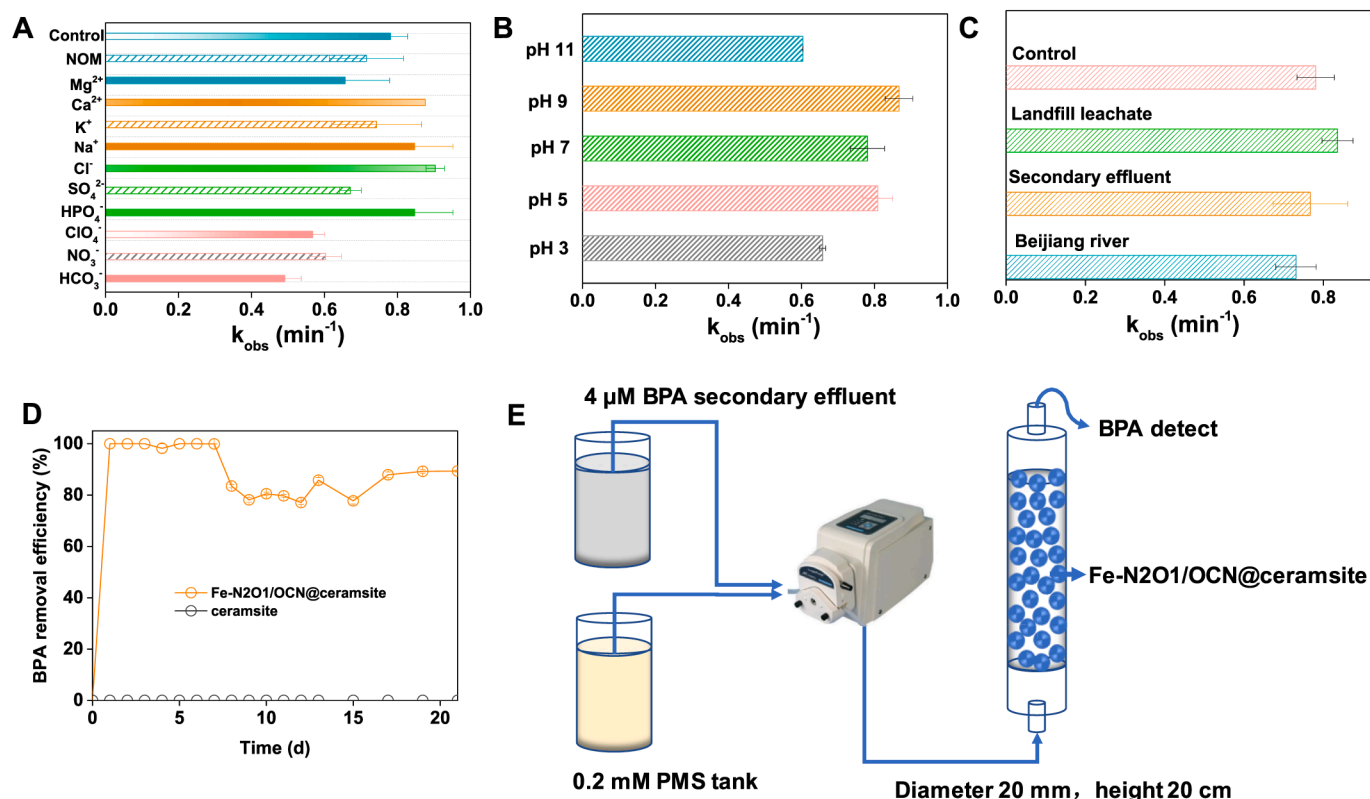


Fig. 5. Evaluation of the practical application of Fe-N2O1/OCN-PMS system. BPA degradation rate in (A) Fe-N2O1/OCN activated PMS system in the presence of different ions with a concentration of 50 mM and in the presence of NOM with a concentration of 20 mgC/L. (B) Effects of pH on BPA degradation in the Fe-N2O1/OCN system. Experimental conditions: [BPA] = 2 μ M, [catalyst] = 0.1 g/L, [PMS] = 0.2 mM. (C) The BPA degradation rate in different natural waters of Fe-N2O1/OCN activated PMS systems. Experimental conditions: [BPA] = 2 μ M, [catalyst] = 0.1 g/L, [PMS] = 0.2 mM. (D) The degradation performance of BPA in Fe-N2O1/OCN@ceramsite in the bench scale. (E) Schematic diagrams of the Fe-N2O1/OCN@ceramsite -catalyzed PMS process at the bench scale.

catalyst exhibits a higher anti-interference capability compared to several advanced catalysts, including Co-N3 [53], FeNx-B-2 [54], Co₃O₄/CeO₂-IM [55], ZnFeMnO₄ [19], Co_{1.1}Mn_{1.9}O₄ [56] and Fe-N-C [45]. These characteristics make the Fe-N2O1/OCN activation of PMS an excellent process for decomposing BPA in real water matrices. The degradation rate of BPA in real waters such as Beijiang River, secondary effluent, and landfill leachate wastewater, was as high as that in ultrapure water (Fig. 5C), ignoring the influence of coexisting matrix. This process demonstrated superior performance compared to the radical-based AOPs, which are greatly affected by the ions and NOM [57].

The cyclic tests confirmed the stability of Fe-N2O1/OCN in degrading BPA, as shown in Fig. S36. In these tests, BPA was periodically added to the system while PMS was only dosed once at the beginning. The degradation rate of BPA decreased as the PMS concentrations diminished. However, the addition of PMS again restored the high degradation rate of BPA. After six cyclic reactions, complete removal of BPA was achieved, surpassing the performance of several state-of-the-art catalysts as indicated in Table S7. The leaching concentration of Fe in the Fe-N2O1/OCN system was about 0.4 mg/L, which was smaller than that in the Fe-N4/OCN system, about 0.7 mg/L (Fig. S38). Furthermore, analysis of Fe-N2O1/OCN before and after use through XRD, FTIR, XPS, and Raman spectra showed no significant variation, indicating the satisfactory structural stability of Fe-N2O1/OCN even after six cyclic reactions (Fig. S39). Furthermore, a flow-through test was also conducted to demonstrate the potential application of Fe-N2O1/OCN in real water treatment. Solutions containing 2 μ M BPA and 0.1 mM PMS flowed through a column filled with Fe-N2O1/OCN loaded ceramsite, consistently removing more than 80% of BPA for over 20 days (Fig. 5D-5E). Therefore, Fe-N2O1/OCN based on surface-activated PMS@complex is an excellent candidate for environmental water treatment. Lastly, the

cost of the bench-scale experiment was estimated to be only 0.055 CNY·L⁻¹ of the secondary effluent (Table S11).

4. Conclusions

In conclusion, we have reported the successful development of a long-lasting and anti-interference AOP system using Fe-N2O1/OCN. This system demonstrates remarkable performance in various water matrices for a continuous operation period of 20 days while maintaining a low cost of only 0.055 CNY·L⁻¹. The exceptional anti-interference and affordability of this system can be attributed to the PAPO pathway for PMS activation. The PMS@Fe-N2O1/OCN complex was formed upon PMS addition, which can be activated by BPA and generate R• and SO₄^{•-} adsorbed on Fe-N2O1/OCN. Both oxidants degrade BPA simultaneously. This process is achieved through the oxygen-coordination iron center, which weakens the coordination between PMS and Fe-N2O1/OCN, thereby preventing its breakage and promoting the formation of the PMS@Fe-N2O1/OCN complex. In this study, an affordable, long-lasting, and anti-interference AOP system has been developed, which will greatly enhance the application of AOPs in wastewater treatment.

CRedit authorship contribution statement

Yanchun Deng: Data curation, Writing – original draft, Writing – review & editing, Formal analysis. **Yu-Xin Ye:** Funding acquisition, Writing – original draft, Writing – review & editing. **Yingge He:** Data curation, Formal analysis. **Jianqiao Xu:** Writing – review & editing. **Zhuofeng Ke:** Data curation, Project administration, Software. **Xinran Zhang:** Funding acquisition, Writing – review & editing. **Gangfeng Ouyang:** Writing – review & editing, Conceptualization. **Xin Yang:**

Funding acquisition, Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2024.149782>.

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